PROJECT REPORT

# WiFi Dig

Router & Network Intelligence Suite

A Python-based desktop utility for exploring network information, diagnosing common connectivity issues, examining nearby wireless network conditions, and reviewing connection performance.

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| **Project version** | Version 2.0 |
| **Project title** | Router & Network Intelligence Suite |
| **Development context** | Developed during an IT/networking internship at AA Network |
| **Technology** | Python desktop application with a Tkinter graphical interface |

Prepared as a concise technical project overview.

# Abstract

WiFi Dig is a desktop network-diagnostic utility developed in Python to bring several practical network-inspection tasks into one graphical application. Rather than relying on separate command-line tools for every check, the application presents network information, local-network discovery, wireless channel observations, router-related details, connection-performance tests, and selected electrical/RF calculations through a single interface. It also supports exporting diagnostic information in common file formats. The project focuses on making preliminary network investigation more accessible and organised. Its readings depend on the computer, operating system, network adapter, available permissions, and external test services; it is therefore a diagnostic aid rather than a replacement for professional network-testing equipment.


# 1. Introduction

Wireless connectivity is now part of everyday study, work, and communication. When a connection becomes slow or unstable, identifying the likely source of the problem can involve checking the computer's network configuration, the local network, the wireless environment, and the quality of the internet connection. These checks are often spread across different utilities and system menus.

WiFi Dig was developed to organise a selection of these checks in one place. The project combines practical computer-networking concepts with a desktop software interface. It is intended to help a user inspect available information, run basic performance checks, and save results for later review.

# 2. Project Objectives

- Create a desktop interface that brings common network-diagnostic functions together.
- Display useful information about the computer's network connection and local network.
- Support local-area network (LAN) device discovery and wireless channel inspection.
- Provide connection-performance indicators such as download and upload speed, latency, jitter, signal strength, and packet loss where supported.
- Include router-related information and selected electrical/RF calculation tools.
- Offer preliminary troubleshooting insights and allow diagnostic data to be exported.

# 3. System Overview

The application is structured as a graphical desktop tool. The user selects a diagnostic area, the program gathers or calculates the relevant information, and the interface presents the result. Some functions rely on information reported by the operating system or network adapter; performance testing uses an internet speed-test service; and calculator functions use supplied inputs. Export options allow selected information to be saved outside the app.

# 4. Main Features and Modules

| Module | What it does | Practical value |
|---|---|---|
| **Network information** | Displays available details about the active network connection and network configuration. | Helps the user understand which connection and network settings the computer is using. |
| **LAN device discovery** | Checks for devices visible on the local network, subject to network configuration and permissions. | Provides a basic view of devices present on the same local network. |
| **Wi-Fi channel analysis** | Examines available wireless-network/channel information reported by the system or adapter. | Can help the user notice channel congestion or overlapping wireless activity. |
| **Wi-Fi performance** | Includes download and upload speed, ping/latency, jitter, signal strength, packet loss, connection status, refresh, and test history where supported. | Collects several indicators that can help describe connection performance. |
| **Router information** | Presents router-related specifications or details available to the application. | Keeps relevant router information alongside other network checks. |
| **Electrical/RF calculators** | Provides calculation tools for selected electrical and radio-frequency values. | Connects networking topics with supporting technical calculations. |
| **Cost estimates** | Calculates estimates using the values and assumptions entered by the user. | Offers a simple way to compare estimated costs for a given scenario. |
| **Export tools** | Saves supported results in TXT, JSON, and CSV formats. | Makes results easier to store, share, and review in other applications. |
| **Troubleshooting insights** | Provides preliminary diagnostic observations based on the information collected. | Helps the user decide which basic checks may be worth performing next. |

Note: The exact information available in individual modules can vary by operating system, wireless adapter, router, network permissions, and internet connection.

# 5. Wi-Fi Performance Measurements

The performance area is designed to present several indicators rather than treating internet speed as the only measure of connection quality. Download speed describes the rate at which data is received; upload speed describes the rate at which data is sent. Ping or latency indicates response time, while jitter describes variation in latency. Packet loss indicates the proportion of test packets that do not reach their destination. Signal strength may be shown in dBm or as a percentage when the adapter and operating system provide that information.

The performance test uses the **speedtest-cli** tool for internet speed measurement. Results are affected by the selected test server, network traffic, router performance, wireless interference, and the computer's connection. Repeated tests may therefore produce different values.

# 6. Technical Design and Workflow

## 6.1 Development technologies

- **Python:** Main programming language used to implement the application logic and diagnostic workflows.
- **Tkinter:** Used to build the desktop graphical user interface and organise the available tools.
- **psutil:** Used for supported system and network-interface information.
- **speedtest-cli:** Used for internet connection speed testing.
- **TXT, JSON, and CSV:** Export formats for storing supported diagnostic results.

## 6.2 Typical user workflow

A typical session begins when the user opens WiFi Dig and selects a module. For a basic connection check, the user can review the displayed network information and then run a performance test. For local-network investigation, the user can use the LAN discovery and wireless channel tools where the computer and network permit them. Router details and the built-in calculators provide additional context. The user can then review the preliminary insights and export supported results for later reference.

## 6.3 Design approach

The project follows a modular utility-based approach: related tasks are grouped into separate sections, while the graphical interface provides a consistent way to access them. This makes the tool easier to navigate than a collection of unrelated commands. Export functionality also separates the act of running a diagnostic from the later task of documenting or comparing its results.

# 7. Testing and Interpretation

The application is intended for practical, preliminary inspection on a computer connected to a network. Useful checks include confirming that network details are populated, verifying that LAN discovery returns devices when the local network allows discovery, comparing repeated speed-test results, and confirming that export files can be opened in their respective formats. Wireless channel and signal information should be interpreted in the context of the adapter and operating system that reported it.

No universal performance result is reported in this document because network measurements depend on the user's live environment. The report therefore describes the implemented functions and the way their outputs should be interpreted, rather than presenting invented benchmark values.

# 8. Limitations

- The application depends on the information exposed by the operating system, network adapter, and available permissions.
- A precise physical Wi-Fi coverage range cannot be reliably calculated from a single signal reading. Signal quality varies with distance, walls, interference, antenna design, and device hardware.
- LAN discovery may not identify every device, particularly when devices are isolated, asleep, firewalled, or on a different network segment.
- Speed-test results are snapshots and can vary with server choice, internet congestion, and local network conditions.
- Router specifications and wireless details may be incomplete when the router or adapter does not expose them to the computer.

- The tool provides preliminary diagnostic guidance; it does not automatically repair network faults or replace specialist network-analysis equipment.

# 9. Future Development

There are several ways the project could be extended. Possible improvements include a clearer history dashboard for comparing tests over time, additional filters for LAN discovery, more detailed wireless-channel visualisation, configurable diagnostic thresholds, and improved report generation. Testing across different routers, wireless adapters, and operating systems would also help establish where the application behaves consistently and where platform-specific differences need to be handled.

# 10. Conclusion

WiFi Dig brings a range of practical network-inspection and calculation features into a single Python desktop application. Its combination of network information, LAN device discovery, wireless channel analysis, connection-performance measurements, router-related details, electrical/RF calculators, cost estimates, troubleshooting insights, and data export demonstrates an applied use of programming in computer networking. The project also highlights an important engineering principle: diagnostic readings must be interpreted with an understanding of the hardware, environment, and measurement limitations behind them.

## Project summary

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| **Project** | WiFi Dig — Router & Network Intelligence Suite |
| **Version** | 2.0 |
| **Application type** | Python desktop diagnostic utility |
| **Interface** | Tkinter graphical user interface |
| **Core areas** | Network inspection, wireless analysis, performance testing, calculations, exports |
| **Export formats** | TXT, JSON, CSV |

**Documentation note:** This report is based on the project features described for WiFi Dig version 2.0. Actual outputs and available fields may differ depending on the computer and network environment.